YaPcore: YaP-Folia Game Authority, Slim Edge Chassis, Native Network Stack, and First-Party Plugin Suite

YapLabs Technical Whitepaper
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> Prefer plain English? See YAPCORE_WHITEPAPER_PLAIN_ENGLISH.md.
> Operator rundown: YAPCORE_WHITEPAPER.md.

Abstract

Minecraft-class game servers traditionally serialize world mutation, plugin callbacks, and much of network I/O onto a single “main” thread, trading simplicity for latency under concurrent load. **YaPcore** ships as a three-layer product:

1. **YaP-Folia** — YapLabs’ maintained fork of PaperMC Folia **26.2** — owns regionized game tick (`lib/yap-folia-26.2.jar`, `folia-jar-source=build`).
2. **YapEngine** — a slim chassis in the YaPcore parent process — owns the public edge: watchdog, Netty traffic/sequencing, Compatibility Bridge, UI sandboxes, and heavy I/O workers (**not** world tick).
3. **YaP Link** — a first-party Velocity-class proxy (`0.6.0-phase6`) — fronts multi-backend networks.

A **SequenceToken** model orders work across chassis streams. Folia-aware first-party plugins use an explicit **SYNC / HEAVY / UI** contract via `YapSched`. The product ships a **CORE+NETWORK** plugin suite (permissions, chat, moderation, playerdata/economy, protect, world, regions, map, …) and an opt-in **GAMEPLAY** tier (vehicles, stacker, knobs, and a full **MMO** stack M0–M7). Dual-stack **Java TCP + Bedrock UDP** is first-party (no Via*/Geyser jars). Stock Fill Folia and Paper + Phase 3 spatial tick remain **legacy / bench** paths only.

This paper describes architecture, concurrency invariants, networking/crossplay, the shipped plugin and data plane, evaluation methodology, and honest product status as of September 2026.

Keywords: game server concurrency; Folia fork; regionized tick; plugin compatibility; Minecraft protocol; dual-stack crossplay; generational ZGC; MariaDB; MMO progression.

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1. Introduction

1.1 Problem

Vanilla and Paper-derived Java Edition servers concentrate authoritative world updates on one thread (~20 TPS). Plugins that perform database or HTTP work on that thread stall ticks. Unsynchronized parallel mutation produces torn chunks, duplicate entities, and inventory races. Network operators still assemble **LuckPerms + EssentialsX + CoreProtect + WorldEdit + WorldGuard + Velocity + Geyser + …** by hand. Bedrock clients require a second transport (UDP) while operators demand a single shared world.

1.2 Contribution

YaPcore contributes:

1. A **three-layer product stack**: YaP-Folia game tick + YapEngine edge/I/O chassis + YaP Link proxy.
2. A **maintained Folia fork** (branding, teleport transactions, optional hot-region budgets / partition) — QUICK_START.md.

3. **SequenceToken** sequencing for ordered handoff across chassis threads — YAPCORE_WHITEPAPER.md.
4. A **Compatibility Bridge** that stages legacy Bukkit mutations onto the game-core drain window (non-product Paper path; best-effort on Folia via Folia APIs + `YapSched`).
5. Dual-stack **Java TCP + Bedrock UDP** ingress with optional shared listen port — first-party code, not Via*/Geyser jars — CROSSPLAY.md.
6. A **three-tier extension model**: Folia-aware plugins (`plugin.yml`), YaP plugins (`yap.yml`), and fine-tune modules (`module.yml`) — PLUGINS.md · MODULES_AND_API.md.
7. A **shipped first-party plugin suite** that replaces the common DIY glue stack for ~90% of survival/network operators — §6.
8. An opt-in **RuneScape-style MMO progression stack** (skills, combat, crafting, quests, abilities, Bedrock UI) — MMO_PHASES.md.

1.3 Non-goals

- **Stock Paper plugins on YaP-Folia** are unsupported (same reality as upstream Folia). Prefer Folia-aware jars or YaP natives — PLUGIN_COMPAT_MATRIX.md.
- YaPcore is **not** a clean-room rewrite of Minecraft; it **forks Folia on purpose**.
- We do **not** claim "faster than Paper/Leaf on every workload." Population cite: **fullcite** (100 active bots + fixtures) under the **ship Folia knob profile** (async-save, hopper budget, MSPT-gated entity/microtick budgets, subregion partition) — yapcore **−12.4%** vs stock Folia (`20260904TshipFc2`); knobs disclosed in bench JSON (`knob_*`). See `YAP_FOLIA_SOAK.md`. **250 keepalive = HOLD-ONLY**.
- Bedrock play-depth is **join/spawn + play-depth smoke green**; some fidelity rows remain partial — CROSSPLAY.md.
- The Compatibility Bridge facade (non-game authority) remains **best-effort stubs** — PAPER_API_COVERAGE.md.

1.4 Audience

Audience	Primary sections
Operators / network owners	§3, §6–10, §13
Plugin authors	§5–7, Appendix A
Systems / engine contributors	§3–4, §11–12
Non-technical readers	Plain English whitepaper

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2. Related work

Paper / Purpur / Pufferfish / Leaf improve the classic Bukkit single-main-tick model (performance and/or knobs). **PaperMC Folia** introduces regionized multithreading for Bukkit-class servers. **YaPcore forks Folia** as **YaP-Folia** rather than shipping upstream Fill as the product jar. Peer Folia forks (e.g. Canvas) typically leave operators to assemble proxy + plugins themselves.

Netty-based proxies (Velocity; YaP Link) separate player routing from world authority. **Geyser / Via*** stacks provide dual-stack crossplay as separate jars; YaPcore embeds Via-class and Geyser-class code in chassis/`crossplay` packages.

YaPcore sits as: **YaP-Folia's game + deterministic YapEngine thread roles + first-party Link + shipped plugin/data plane**, rather than "DIY Folia + Velocity + ten community plugins." Comparison matrix: YAPCORE_WHITEPAPER.md.

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3. Architecture

3.1 Three layers (product)

```
 Clients (JE TCP / BE UDP)
           |
           ▼ optional
    YaP Link :25565  ──forwarding──►  YaPcore Via edge :25566
           |                                     |
           ▼                                     ▼
  YapEngine chassis (YaPcore parent)      YaP-Folia child JVM
    Traffic · Bridge · UI · Heavy I/O        **GAME TICK** (regions)
                                             folia-kernel/ + lib/yap-folia-*.jar

Legacy (benches only): game-authority=paper + Phase 3 on chassis
```

Layer	Game tick?	Role
YaP Link	No	Proxy JVM — multi-backend routing, compression, Link plugins (`0.6.0-phase6`)
YapEngine chassis	No	Edge, bridge, UI/Heavy I/O, telemetry, dual-stack gateway
YaP-Folia	**Yes**	Region thread pool in embedded JVM (`lib/yap-folia-*.jar`)

Default product properties:

```
game-authority=folia
folia-embed=true
```

```
folia-version=26.2
folia-jar-source=build
folia-teleport-transactions=true
paper-phase3-tick-bridge=false
paper-phase3-nms-tick=false
```

3.2 Chassis channel matrix (T1–16)

ID	Role	Responsibility (v2.0 / product)
1	Controller	Watchdog, recovery, process health
2	Traffic Cop	Ingress shaping, SequenceToken assignment
3–6	Worker quads	Sequenced bridge/plugin tasks; **legacy Phase 3 NMS tick on Paper benches only**
7	Chunk Sync DLM	Deferred lease / chunk ownership (**Paper Phase 3 legacy**)
8	Boundary Arbitrator	Cross-quadrant handoff (**Paper Phase 3 legacy**)
9	Compatibility Bridge	Legacy SYNC mutation queue → YaP-Folia region APIs
10–11	UI sandbox	Menu polish, click routing
12–15	Heavy I/O	DB, HTTP, files, proxy sync
16	Telemetry	Metrics / JFR hooks

See YAPCORE_WHITEPAPER.md.

3.3 Sequencing

Each logical stream obtains a `SequenceToken` carrying a per-stream sequence and a global identifier. Strict ordered queues refuse out-of-order commits within a stream while allowing cross-stream parallelism — YAPCORE_WHITEPAPER.md.

3.4 Spatial model

YaP-Folia indexes world interest by region and runs authoritative tick on a dynamic region thread pool. Optional YaP patches add teleport transactions (default **on**), entity tick budgets, async chunk save, scoreboard SWMR, and experimental subregion partition (default **off**) — §4. **YapEngine chassis quads (T3–6)** route sequenced bridge/plugin work; they do **not** replace YaP-Folia game tick. Legacy Paper Phase 3 used quads + T7/T8 for interior NMS tick (**benches only**).

3.5 Memory & GC posture

Production launch scripts prefer **Generational ZGC** with optional **NUMA** pinning — TUNE.md. **Java 25+** is required.

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4. YaP-Folia fork

YaPcore does **not** ship stock PaperMC Folia as the product game jar. Upstream pin lives in `vendor/folia/UPSTREAM.lock`; ordered patches in `vendor/folia/patches/`.

Patch	Purpose	Default
`0000`	Branding → YaP-Folia	always
`0001`	Cross-region teleport PREPARE/COMMIT/CONFIRM	**on**
`0010`	Async chunk save (Moonrise flush off region thread)	**on** (ship)
`0011`	Scoreboard SWMR	**on** (ship)
`0012` / `0023`	Smart Mob AI tick budget (MSPT-gated)	**400** (ship); 0=off
`0013`	Region pool metrics / microtick knobs	microtick **8 ms** (ship)
`0014`–`0019` / `0024`	Subregion partition + harden	**on** (ship; hysteresis)
`0022`	Hopper BE transfer budget	**64** (ship)

Build: `./scripts/build-yap-folia.sh` → `lib/yap-folia-26.2.jar`.
Docs: YAP_FOLIA_SOAK.md · REAL_GAINS.md · QUICK_START.md.

Stock Folia fallback: `folia-jar-source=fetch` + `./scripts/fetch-folia.sh` (bench / comparison only).

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5. Plugin concurrency contract

Pool	Allowed work	Forbidden
SYNC (region / entity / global via `YapSched`)	Block / inventory / teleport / world	Blocking DB / HTTP
HEAVY	JDBC, HTTP, disk, messaging	Direct block set / `openInventory` without hop
UI	Menu animation, polish	Authoritative world writes

Rules for every first-party jar:

- `folia-supported: true` in `plugin.yml`
- World/block/entity work via `com.yapcore.sched.YapSched` (or Folia region APIs)
- Shared persistence via **YaPDB** when MariaDB is required
- Public APIs registered on Bukkit `ServicesManager`

Operator layout: all jars drop into **`plugins/`** (symlinked into `folia-kernel/plugins`). Fine-tune packaging modules drop into **`modules/`**. See PLUGINS.md · PLUGIN_COMPAT.md.

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6. Shipped first-party plugins

Sources live under `yap-first-party/`. Install tiers:

Tier	Gradle	Audience
CORE+NETWORK	`gradle installProductDefaults`	Every release
GAMEPLAY	`gradle installGameplayDefaults` or `-PyapGameplay=true`	Opt-in survival / MMO
Both	`gradle installAllProductDefaults`	Full product box
Dist	`gradle assemblePluginDist`	`build/dist/yap-plugins/{core-network,gameplay,api,modules}/`

6.1 CORE + NETWORK (default product)

Jar	Plugin	Role
`yap-db.jar`	YaPDB	Shared MariaDB Hikari pool
`yap-perms.jar`	YaPPerms	Groups, tracks, prefixes (`/yapperm`, `/promote`)
`yap-playerdata.jar`	YaPPlayerData	Cross-server sync, auth, economy, homes/warps/kits/mail, **chest shops**, **AH**, claims
`yap-moderation.jar`	YaPModeration	Ban / mute / warn / kick + history
`yap-essentials.jar`	YaPEssentials	Essentials-class QoL (`/spawn`, `/tpa`, `/fly`, `/vanish`, …)
`yap-chat.jar`	YaPChat	Channels, PM, filter, staff chat
`yap-packs.jar`	YaPPacks	Multi resource-pack push
`yap-floodgate.jar`	YaPFloodgate	Bedrock identity without Floodgate jar
`yap-placeholderapi.jar`	PlaceholderAPI	Clip-compatible PAPI (eCloud intentionally stubbed; drop expansions locally)

Jar	Plugin	Role
`yap-plugin-compat.jar`	YaPPluginCompat	1.20–1.21 → 26.2 back-compat status
`yap-pregen.jar`	YaPPregen	Chunk pre-generator
`yap-folia-bridge.jar`	YaPFoliaBridge	Folia surface / scheduler smoke
`yap-protect.jar`	YaPProtect	CoreProtect-class audit / rollback
`yap-world.jar`	YaPWorld	FAWE-class edit (masks, brushes, schems) + world mgmt
`yap-regions.jar`	YaPRegions	WorldGuard-class cuboid flags
`yap-npcs.jar`	YaPNpcs	Quest NPCs + dialogue
`yap-tab.jar`	YaPTab	Tab list / header / footer / sidebar
`yap-discord.jar`	YaPDiscord	Discord webhooks + relay
`yap-guard.jar`	YaPGuard	Lightweight anti-cheat heuristics
`yap-lagguard.jar`	YaPLagGuard	Per-chunk lag governor
`yap-map.jar`	YaPMap	Web flat map (Leaflet + PNG tiles)
`yap-factions.jar`	YaPFactions	Factions overlay on playerdata claims
`yap-bedrock-ui.jar`	YaPBedrockUI	Bedrock `FormService` bridge

6.2 GAMEPLAY (opt-in)

Jar	Plugin	Role
`yap-vehicles.jar`	YaPVehicles	Real vehicle mechanics (not minecarts)
`yap-stacker.jar`	YaPStacker	PDC mob / item / spawner stacker
`yap-gameplay-knobs.jar`	YaPGameplayKnobs	Purpur-class mob encyclopedia
`yap-skills.jar`	YaPSkills	13 RS-style skills + `/skills` GUI
`yap-combat.jar`	YaPCombat	Custom PvE combat, gear, food, potions, spells, prayer
`yap-crafting.jar`	YaPCrafting	Recipes, stations, `/sell`
`yap-mmo-content.jar`	YaPMmoContent	Quests v2, bosses, skill areas, hiscores
`yap-abilities.jar`	YaPAbilities	Config-driven ability engine (230+)
`yap-mechanics.jar`	YaPMechanics	Stamina, resource nodes, farming

Jar	Plugin	Role
`yap-games.jar`	YaPGames	Minigames (arenas, queue, duels)
`yap-guilds.jar`	YaPGuilds	MMO guilds
`yap-mmo-bedrock.jar`	YaPMmoBedrock	Bedrock MMO forms UI

MMO milestones M0–M7 are shipped — MMO_PHASES.md. Combat skill XP is owned by **YaPCombat** when loaded; YaPSkills provides a vanilla-damage fallback when combat is absent.

6.3 APIs & modules

Nineteen `yap-*-api` jars under `yap-first-party/api/` for soft-depend authors. Fine-tune modules under `modules/` declare `provides` / `requires` and write `FINE_TUNE.txt` pointers at real config knobs — they are **packaging**, not alternate game engines. Games packaging modules (`yap-games-module`, FFA, duels) install with GAMEPLAY / `installFineTuneModules`.

6.4 What first-party plugins intentionally replace

Community staple	YaP native
LuckPerms	YaPPerms
EssentialsX (QoL)	YaPEssentials (+ playerdata for data/economy)
CoreProtect	YaPProtect
WorldEdit / FAWE-class ops	YaPWorld (Folia-safe; stock FAWE not used)
WorldGuard	YaPRegions (+ playerdata claims)
TAB / scoreboard	YaPTab
DiscordSRV (MVP)	YaPDiscord
Dynmap-class	YaPMap
PlaceholderAPI	Bundled Clip-compatible engine
Floodgate	YaPFloodgate
Velocity	YaP Link
Via* + Geyser jars	Chassis dual-stack

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7. Data plane

7.1 YaPDB

`yap-db.jar` exposes a shared Hikari pool. Prefer `use-shared-yapdb: true` in consumers. Setup: `./scripts/db/ensure-db.sh --server-id lobby` — YAPDB.md · MARIADB.md.

7.2 YaPPlayerData

Cross-server inventory / XP / vitals sync, session lock (always on), optional offline `/login`, economy (`/bal` `/pay` + Vault), homes/warps/kits/mail, claims + tax, NPC traders (opt-in).

Default feature flags (September 2026):

Feature	Default
Economy	**on**
Homes / warps / kits / mail / claims	**on**
Chest shops (`/shop`)	**on**
Auction house (`/ah`)	**on**
Jobs	**off** (keep off when YaPSkills is enabled — avoids double mining payouts)
NPC traders	**off** (opt-in)

Smoke: . Docs: PLAYERDATA.md.

7.3 Permissions

Native ranks/groups/tracks — PERMISSIONS.md. Soft integration with TAB prefixes and dashboard ranks APIs.

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8. YaP Link

First-party **Velocity-class** proxy — **not** a Velocity fork. Phases **0–6 shipped** (`0.6.0-phase6`): passthrough ping, forced hosts, health failover, chat relay, system chat, Link plugin loader, edge rate limits, metrics hooks, two-backend smoke.

Link plugins (`link-plugin.json`): chat-bridge, mod-sync, server-selector, tab-bridge, discord. Docs: YAP_LINK.md · YAP_LINK_NATIVE.md. Stock Velocity remains optional for migration — VELOCITY.md.

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9. Networking & crossplay

Path	Notes
Java Edition	Framed Netty; Via-class remapping in chassis (`com.yapcore.protocol.via*`). Target protocol ~**776** (26.2). JE floor **1.20.2+**.
Bedrock	UDP path; Geyser-class hub in `com.yapcore.crossplay*` — **no Geyser jar**.
Shared listen	Optional same port for JE+BE (`shared-listen-port=true`).
YaP Link	Optional JE front + optional Bedrock UDP forwarder.
Packs HTTP	Default pack `resourcepacks/yapcore-default.zip` on `:8081`.
Publicity	Domain / SRV / nginx + Cloudflare — NETWORKING.md.

Same-machine clients must use `127.0.0.1` (hairpin NAT) — NGINX_AND_LOCALHOST.md.

Phase 4 dual-stack **join DoD is green**; play-depth smoke green; remaining fidelity rows tracked in CROSSPLAY.md.

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10. Ops surface

Surface	Detail
Config	`config/server.properties` + hub under `config/`
Control GUI	Desktop Swing panel
Web dashboard	Token-auth browser UI `:8080` — WEB_DASHBOARD.md
Crash dumps	`logs/crashes/crash-<ts>-<kind>.log`
Release	`gradle assembleRelease` → `build/dist/yapcore-release/{linux,windows}/`
Windows	Parity launchers — WINDOWS.md

Dashboard Phase 8 polish remains a backlog item — see WEB_DASHBOARD.md.

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11. Evaluation methodology

Unit tests (JUnit) cover plugin and API behavior. Operators validate with a local boot: `./scripts/seed-defaults.sh` then `./scripts/start.sh --fg`.

Metrics: region MSPT p99, bridge queue depth, join success by protocol version, HEAVY pool saturation, MariaDB pool health.

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12. Threats to validity

- Folia (and YaP-Folia) reject stock single-thread Paper plugins — operator education required.
- Protocol version sprawl requires continuous registry/packet maintenance.
- NUMA/ZGC gains are hardware-dependent.
- Bedrock play-depth parity still lags Java for some packets / UI surfaces.
- YaP-Folia patch rebase risk when refreshing `UPSTREAM.lock`.
- Phase 3 spatial edge cases apply only on the **legacy Paper** path.
- Soft-depend plugin names are **case-sensitive** (`YaPPlayerData`); integrations must match `plugin.yml` `name:` exactly.

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13. Product status (September 2026)

Area	Status
YaP-Folia product path	**Default** (`folia-jar-source=build`)
YapEngine slim chassis	**Always on**
YaP Link phases 0–6	**Shipped** (`0.6.0-phase6`)
Phase 3 Paper spatial	**Complete as code — retired as product default**
Phase 4 dual-stack join DoD	**Green**
CORE+NETWORK plugins	**Shipped**
GAMEPLAY + MMO M0–M7	**Shipped** (opt-in install)
PlayerData shops + AH	**On by default** (jobs remain off)
Fair population MSPT gate	**Citeable** — fullcite 100 bots, yapcore −12.4% vs stock Folia (**ship knobs** disclosed in JSON)
Dashboard Phase 8 full tabs	**Partial / roadmap**
PAPI eCloud	**Intentionally stubbed**
Stock Paper jars on Folia	**Unsupported**

Ops polish backlog: WEB_DASHBOARD.md.

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14. Conclusion

YaPcore demonstrates a practical decomposition for Minecraft-class servers: **YaP-Folia** owns regionized game tick; **YapEngine** owns a slim edge/I/O chassis with an explicit plugin pool contract; **YaP Link** owns multi-backend routing; and a **first-party plugin + MariaDB data plane** replaces the usual DIY glue stack. Opt-in GAMEPLAY and MMO tiers extend the same Folia-safe patterns for vehicles and RS-style progression.

Future work emphasizes deeper dual-stack fidelity, hot-region partition soak under load, dashboard ops completion, and continued Folia upstream rebase hygiene.

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15. References

1. Minecraft Wiki — *Java Edition protocol*.
2. OpenJDK — *Generational ZGC*.
3. Netty project.
4. PaperMC Folia — regionized threading for Bukkit servers.
5. YapLabs — YaP-Folia patches (`vendor/folia/patches/`), YapEngine chassis notes, YaP Link native suite.
6. YapLabs docs — YAPCORE_WHITEPAPER.md, YAPCORE_WHITEPAPER.md, MMO_PHASES.md.

Appendices

Appendix A — Document map

Audience	Start here
Non-tech readers	Plain English whitepaper, YAPCORE_WHITEPAPER_PLAIN_ENGLISH.md
Operators	QUICK_START.md, QUICK_START.md, PLUGINS.md
Plugin authors	PLUGINS.md, MODULES_AND_API.md, MODULES_AND_API.md
Engine contributors	YAPCORE_WHITEPAPER.md, YAPCORE_WHITEPAPER.md, QUICK_START.md
Network / crossplay	YAP_LINK.md, CROSSPLAY.md, CROSSPLAY.md
Data	YAPDB.md, PLAYERDATA.md, PERMISSIONS.md
MMO	MMO_PHASES.md, MMO_SKILLS.md, MMO_COMBAT.md
Comparison	YAPCORE_WHITEPAPER.md

Appendix B — Quick install

```
# Java 25+
./scripts/build-yap-folia.sh          # → lib/yap-folia-26.2.jar
./scripts/db/ensure-db.sh --server-id lobby
gradle installProductDefaults         # CORE+NETWORK → plugins/
# optional:
gradle installGameplayDefaults        # vehicles + stacker + MMO …
gradle assembleRelease
./scripts/start.sh --fg
# multi-backend:
./scripts/start-yap-link.sh
```

Appendix C — Citation

```
@techreport{yapcore2026sixteen,
  title       = {YaPcore: YaP-Folia Game Authority, Slim Edge Chassis, Native Network Stack,
and First-Party Plugin Suite},
  author      = {{YapLabs}},
  institution = {YapLabs},
  year        = {2026},
  month       = sep,
  number      = {YAP-WP-16T-001},
  note        = {Technical whitepaper, YaPcore 0.3}
}
```

Appendix D — Changelog (whitepaper)

Ver	Date	Notes
0.2	Aug 2026	Three-layer architecture; chassis T1–16; Link phase6; Folia product path
0.3	**Sep 2026**	Comprehensive plugin catalog; MMO M0–M7; data plane (shops/AH defaults); Link/crossplay/ops status; evaluation smokes; honest roadmap gaps

Appendix E — License

YaPcore first-party code is **GNU GPLv3** (or later). See root `LICENSE` and LICENSING.md. YaP-Folia derivatives inherit Folia/Paper GPLv3 obligations.